

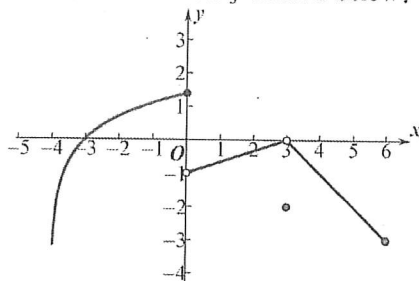
Maio

1.5 函数的连续性(Continuity)

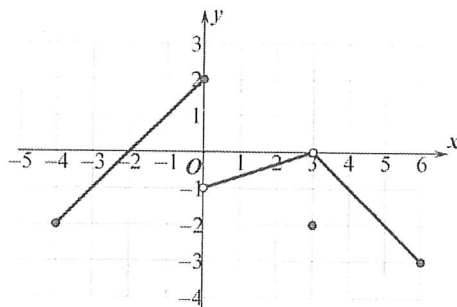


Multiple Choice Questions

- At what point is the removable discontinuity for the function $y = \frac{x^2 + 5x - 24}{x^2 - x - 6}$?
 (A) $x = -8$ (B) $x = 3$ (C) $x = -2$ (D) $x = 3$ and $x = -2$
 - On which of the following intervals is $y = \frac{1}{2\sqrt{x}}$ not continuous?
 (A) $(0, +\infty)$ (B) $(0, 2)$ (C) $[0, \infty)$ (D) $[1, \infty)$
 - Which of the following points is not a point of discontinuity of $y = \frac{1}{\sqrt{x-2}}$?
 (A) 2 (B) 0 (C) -1 (D) 2.01
 - Which of the following statements about the function $f(x) = \begin{cases} 3x, & 0 < x < 1 \\ 1, & x = 1 \\ -x + 4, & 1 < x < 2 \end{cases}$ is true?
 (A) $f(x)$ is continuous at $x = 1$.
 (B) $\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$
 (C) $\lim_{x \rightarrow 1} f(x) = 1$
 (D) $\lim_{x \rightarrow 1} f(x) \neq f(1)$
- Questions 5–6 are based on the function f shown below:



- The function f has a removable discontinuity at
 (A) $x = -4$ (B) $x = 0$ (C) $x = 3$ (D) $x = 6$
- The function f has a jump discontinuity at
 (A) $x = -4$ (B) $x = 0$ (C) $x = 3$ (D) $x = 6$
- The figure below shows the graph of the function f . Which of the following statements is true?



I. $\lim_{x \rightarrow 0^+} f(x) = f(0)$ II. $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$ III. $\lim_{x \rightarrow 3} f(x) = f(3)$.

- (A) II only (B) III only (C) I and II only (D) II and III only

8. For what value of m is $f(x) = \begin{cases} \frac{x^2 - 7x + 10}{m(x-5)}, & x \neq 6 \\ m, & x = 6 \end{cases}$ continuous at $x=6$?

$\frac{x^2 - 7x + 10}{m(x-5)} = 6$

- (A) $m=2$ (B) $m=\pm 2$
(C) $m=4$ (D) There is no such value of m .

9. Find a value for a so that the function $f(x) = \begin{cases} 4 - x^2, & x < -1 \\ ax^2 - 1, & x \geq -1 \end{cases}$ is continuous

everywhere.

- (A) $a = -1$ (B) $a = 0$ (C) $a = 4$ (D) $a = 6$



Free Response Questions

10. For what value of a is the function $f(x) = \begin{cases} 3x^2 - 11x + 4, & x \leq 4 \\ a^2x^2 - 2x - 1, & x > 4 \end{cases}$ continuous at $x=4$?

$3(4)^2 - 11(4) + 4 = 44 - 44 = 0$

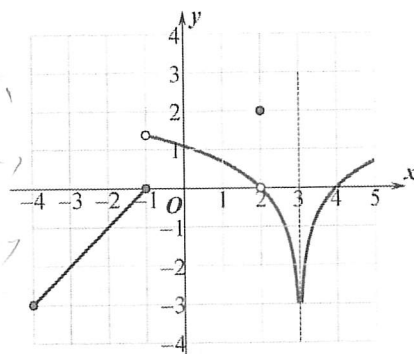
$16a^2 - 9 \geq 0 \quad a^2 \geq \frac{9}{16} \quad a = \pm \frac{3}{4}$

11. Given the graph of $f(x)$ below.

(a) Find all kinds of discontinuous points of $f(x)$.

(b) Find $\lim_{x \rightarrow -1} f(x)$

(c) Is $f(x)$ continuous at $x=2$? Explain why.



$x = -1$, jump discontinuity

$x = 2$, removable discontinuity

$x = 3$, infinite discontinuity

C, IV, because at exactly 2,

the value of x is not the same with 2^+ and 2^-